TEAM HAIRLOSS

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AUTO-WARGAME

Malaysia Policy Simulation Engine

Powered by Z.AI ILMU-GLM-5.1 · UM Hackathon 2026

AI SYSTEMS & AGENTIC WORKFLOW AUTOMATION

6	GLM	4	∞
WORKFLOW STAGES	CENTRAL REASONING	MALAYSIA SCENARIOS	HYPOTHESES POSSIBLE

THE PROBLEM

How do policymakers and businesses navigate complex, cascading decisions?

⚡ Fragmented Analysis

Teams rely on isolated spreadsheets and siloed reports — no system connects economic, industry, and behavioural data together.

🔄 Manual & Slow

Policy impact analysis takes weeks of human coordination across departments, ministries, and data sources.

🌀 Can't Handle Ambiguity

Real-world policy questions are vague and multi-layered. Traditional tools fail when inputs are incomplete or hypothetical.

📉 No Chain-Reaction Modelling

Existing tools show first-order effects only. Second, third, and fourth-order ripple effects across industries go unmodelled.

OUR SOLUTION

Turn a vague policy hypothesis **into a full multi-dimensional impact report** — in minutes.

🧠 Agentic GLM Core

GLM acts as command centre — not a chatbot. It decomposes, reasons, orchestrates, and synthesises autonomously.

🔄 Multi-Step Orchestration

6 chained reasoning stages fire sequentially. Each feeds context into the next for coherent, deep analysis.

📡 Malaysia-Specific

Built around BNM, DOSM, MOF data sources. RON95, OPR, flood chains, AI labour displacement — real local scenarios.

⚙️ Edge Case Resilient

Missing data? GLM automatically falls back to historical proxies (e.g. 2014 subsidy removal data) and flags uncertainty.

🎯 Game Theory Layer

Simulates rational actors: Government, SMEs, B40 households, BNM, opposition — Nash equilibria identified automatically.

📊 Structured Output

Generates executive impact reports with regional differentiation, policy levers, and confidence ratings.

HOW IT WORKS

GLM-Orchestrated 6-Stage Simulation Pipeline

01
INPUT
Vague policy hypothesis entered by user

02
DECOMPOSE
GLM breaks hypothesis into 6 structured subtasks

03
ECONOMIC INTEL
Baseline indicators retrieved; data gaps identified with fallbacks

04
CASCADE MODEL
5-layer industry ripple effect mapped (direct → structural)

05
GAME THEORY
7 rational actors simulated; Nash equilibria computed

06
EDGE CASES
Ambiguity resolved; confidence calibrated; black swans flagged

07
REPORT
Multidimensional impact assessment with regional differentiation

⚡ **ILMU-GLM-5.1 acts as the central reasoning engine connecting and orchestrating ALL stages above**

Remove GLM → the entire pipeline collapses. No decomposition. No reasoning. No report.

`Node.js + Express` `OpenAI-Compatible API` `Z.AI ILMU-GLM-5.1` `Vanilla JS Frontend`

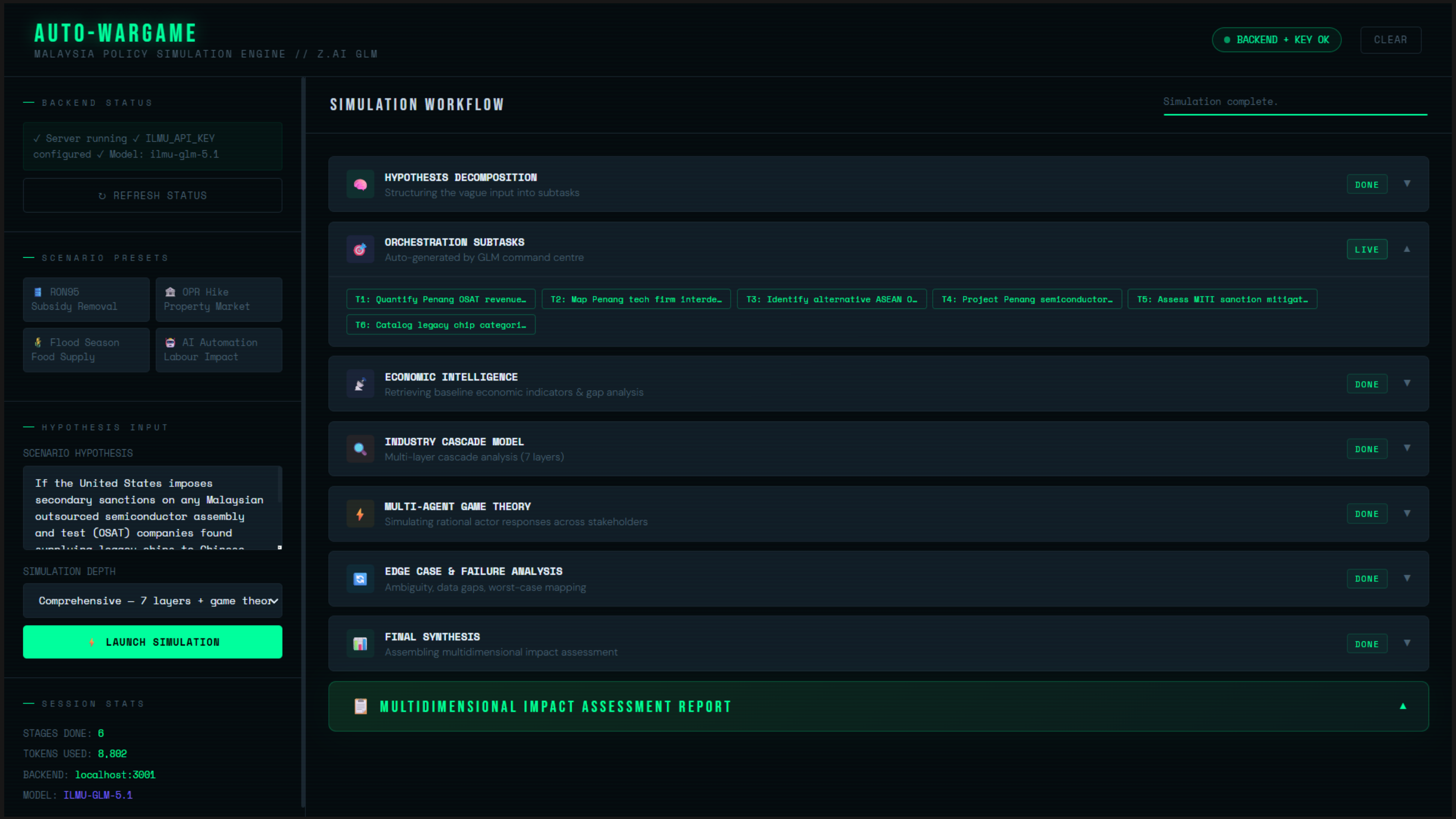
PROTOTYPE DEMO

LIVE SCENARIO: *"If Malaysia eliminates RON95 subsidies next month, what chain reactions trigger in the logistics industry?"*

01 DECOMPOSE	6 subtasks: price impact, supply chain, SME viability, consumer behaviour, regional variation, policy options
02 ECONOMIC	RON95 = 30% of logistics fuel cost. 2014 precedent: +8.5% CPI spike. Fallback: BNM 2022 inflation data used for projection
03 CASCADE	L1: Logistics SMEs hit. L2: E-commerce & fresh food supply disrupted. L3: Urban consumers substitute, rural B40 absorbs cost
04 GAME THEORY	Govt dominant strategy: staged removal + cash transfer. SMEs: lobby + pass-through pricing. B40: demand welfare expansion
05 EDGES	Black swan: global oil spike concurrent. Data gap: future freight volume → extrapolated from 2019-2023 trend. Confidence: MEDIUM

→ Final output: Full executive impact report with regional breakdown, 3 policy response options, and confidence rating

USER INTEFACE



WHY GLM IS THE ENGINE

Meeting the core judging criterion: GLM must be irreplaceable

CAPABILITY	WITH GLM	WITHOUT GLM
Unstructured Input	Interprets any vague hypothesis	Requires structured form input
Subtask Generation	Dynamically creates 6 subtasks per scenario	Pre-hardcoded tasks only
Multi-Step Reasoning	Chains 6 stages with context carry-over	No context — isolated outputs
Edge Case Handling	Auto-detects gaps, selects fallback data	Crashes or returns null
Game Theory Sim	Simulates 7 actors, computes equilibria	Not possible
Final Report	Synthesises all stages into structured output	No synthesis capability

BUILT FOR MALAYSIA

4 real-world Malaysian policy scenarios with local economic context baked in

RON95 Subsidy Removal

Logistics cost cascade, urban vs rural impact, SME survival rates across Peninsular & East Malaysia

OPR Rate Hike

Klang Valley property market impact, first-time buyer vs investor differential analysis

Flood Season Supply Chain

Kelantan/Terengganu/Pahang simultaneous flooding — food security & retail price disruption

AI Labour Displacement

30% automation acceleration — sector-by-sector job creation vs destruction, skill-level differentiation

Data references: Bank Negara Malaysia · DOSM · Ministry of Finance · Ministry of Transport · data.gov.my

EDGE CASE HANDLING

Real-world constraints: ambiguity, missing data, process failures — all handled

📭 MISSING DATA	GLM identifies the gap and automatically selects a historical proxy dataset (e.g. 2014 fuel subsidy removal data used when future freight volumes are unavailable)
🌀 VAGUE HYPOTHESIS	GLM's decomposition stage extracts structured subtasks from any ambiguous natural-language input — no rigid form required
⚠️ API FAILURE	Each stage has independent error handling **(Built-in LLM Hallucination Catch (JSON Fallback))**. If a sub-call fails, the system logs the failure and continues with a fallback response
🦢 BLACK SWAN EVENTS	The edge case analysis stage explicitly models low-probability, high-impact disruptions and flags them in the final report
⚖️ CONFLICTING DATA	GLM calibrates confidence levels per finding (HIGH / MEDIUM / LOW) and surfaces contradictions rather than silently resolving them

FUTURE ROADMAP

PHASE 1

Live Data Integration

- Connect to data.gov.my open API
- Real-time BNM inflation feeds
- Ministry of Transport freight volume data
- Live news sentiment scraping

PHASE 2

Enhanced Simulation

- Agent-based economic modelling
- Probabilistic outcome distributions
- Cross-scenario comparison engine
- Bahasa Malaysia language support

PHASE 3

Enterprise Platform

- PDF report export for policymakers
- Ministry collaboration workspace
- Historical scenario database
- API for third-party integration

AUTO-WARGAME

Turn uncertainty into strategy.

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6	**7**	**4**	**100%**
Workflow Stages	Simulated Actors	Local Scenarios	GLM-Orchestrated

github.com/your-team/auto-wargame